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BCS-3B

22-F3289

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DS LAB-1 (remaining q-5)

Code:

```
#include<iostream>
#include<cstdlib>
using namespace std;

class DynamicImplementations{
public:
    int *arr;
    int size;
    int pos;

    DynamicImplementations(){
        size = 10;
        arr = new int [size];
        pos = -1;
        for(int i= 0 ; i<size ; ++i){
            arr[i]=0;
        }
    }

    DynamicImplementations(int size , int pos = 0){
        this->size=size;
        this->pos=pos;
        arr=new int[this->size];
        for(int i= 0 ; i<size ; ++i){
            arr[i]= rand() % 100 +1;
        }
    }
}
```

```

}

void printList(){
    for(int i = 0 ; i<size ; ++i){
        cout<<arr[i]<<" ";
    }
}

int searchElement(int x){
    for(int i = 0 ; i<size ; ++i){
        if (x == arr[i]){
            cout<<"number is found at "<<i+1<<" index \n";
            return i;
        }
    }
    return 0;
}

void insertElement(int x){
    int *temp;
    size+=1;
    temp = new int [size];
    for(int i = 0 ; i<size-1 ; ++i){
        temp[i]=arr[i];
    }
    temp[size-1]=x;
    arr = new int [size];
    for(int i = 0 ; i<size ; ++i){
        arr[i] = temp[i];
    }
    cout<<"its added !";
    delete[] temp;
}

bool isEmpty(){
    for(int i= 0 ; i<size ; ++i){
        if(arr[i]!=0){
            return false;
        }
    }
    return true;
}

```

```

    bool isFull(){
        for(int i= 0 ; i<size ; ++i){
            if(arr[i]==0)
                return false;

        }
        return true;
    }

    ~DynamicImplementations(){
        delete [] arr;
    }
};

int main()
{
    DynamicImplementations a(5);
    a.printList();
    a.insertElement(2);
    a.printList();
    if(a.isEmpty()){
        cout<<"\nis empty\n";
    }
    else
        cout << "Not empty" << endl;
    if(a.isFull()){
        cout<<"\nis full\n";
    }
    else
        cout << "Not full" << endl;

    return 0;
}

```